KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2017
Department of Computer Science and Engineering
CSE 4127
Computer Vision and Image Processing

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Relate and differentiate between computer vision and image processing. Explain the possible challenges one has to face in solving a computer vision problem. (10)
   b) Define convolution. Show by using an example that $f \star e = f$, where f is an image, e is a unit impulse, and $\star$ is a convolution operator. (10)
   c) Sharp the following image using unsharp mask. Use a 3×3 Gaussian blurring kernel with $\sigma = 1$ and do zero padding where necessary. (15)

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 5 & 2 & 5 & 5 \\ 1 & 1 & 3 & 6 \\ 2 & 4 & 6 & 7 \end{bmatrix}$$

2. a) What is the necessity of transforming an image into frequency domain? Let, $F(u, v)$ is a [M, N] point Discrete Fourier Transform (DFT) of an image defined by $f(x, y)$. Then show that $F(u, v) = F(u + pM, v + qN)$, where p, q are arbitrary constant. (10)
   b) Describe Canny edge detection algorithm in detail. (12)
   c) Design a Butterworth Lowpass Filter (BLPF) to smooth an image. How can you choose the cut-off frequency D_0 to pass α percent of image power? Also explain why the lower order BLPF is more preferable than higher order one. (13)

3. a) Define order statistic filter. Use a 3×3 alpha-trimmed mean filter with $d = 4$ to filter noises from the image presented below. (13)

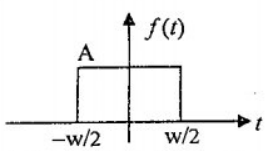
$$\begin{bmatrix} 2 & 3 & 5 & 8 & 7 \\ 8 & 12 & 4 & 6 & 3 \\ 5 & 7 & 6 & 3 & 9 \\ 4 & 1 & 4 & 8 & 3 \\ 6 & 7 & 8 & 14 & 17 \end{bmatrix}$$

   b) What is periodic noise? How can you remove such noise from an image? (05)
   c) Explain inverse filtering process to restore a noisy image. What is the shortcoming of inverse filter? How does Wiener filter overcome such shortcoming? (12)
   d) What is image segmentation? Mention some applications of it. (05)

4. a) Find a global threshold for the following image. What are the problems of using a single global threshold to segment an image? How can these be resolved? (13)

$$\begin{bmatrix} 1 & 2 & 1 & 0 & 12 & 14 \\ 2 & 3 & 4 & 8 & 13 & 12 \\ 9 & 2 & 3 & 15 & 13 & 7 \\ 3 & 2 & 2 & 10 & 11 & 13 \\ 1 & 2 & 0 & 5 & 12 & 14 \\ 2 & 3 & 6 & 8 & 15 & 13 \end{bmatrix}$$

   b) Perform Fourier transform of the function shown in the following figure. The transformation equation is given by $F(u) = \int_{-\infty}^{\infty} f(t) e^{-j2\pi ut} dt$. (08)



   c) Define motion field and optical flow. Explain the Lucas-Kanade optical flow estimation method stating their key assumptions. (14)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is a digital image? What are the stages through which an image passes in a image processing system? Briefly explain each stage. (10)
   b) Briefly discuss about Weber law and optical illusions. (09)
   c) What is mean by sampling and quantization? Write down the transformation matrices of image Scaling, Rotation and Translation. (08)
   d) Define image compression. Briefly describe any one of the image compression techniques. (08)

6. a) Define bit plane slicing for piecewise linear transformation and histogram equalization. Calculate each plane for the following 4-bit image. (10)

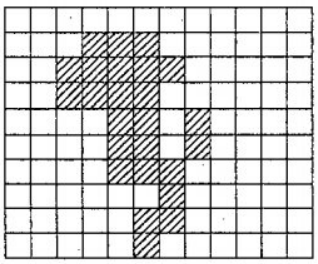
$$\begin{bmatrix} 11 & 13 & 3 & 5 & 8 \\ 8 & 6 & 0 & 11 & 10 \\ 9 & 0 & 2 & 9 & 7 \\ 7 & 12 & 6 & 13 & 14 \\ 3 & 4 & 10 & 0 & 2 \\ 8 & 0 & 15 & 5 & 1 \end{bmatrix}$$

   b) Consider the following 5×5 image with grayscale range of [0, 9]. (17)

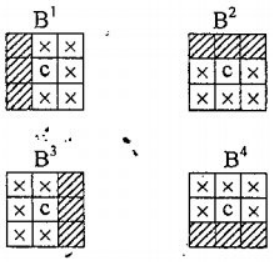
$$\begin{bmatrix} 0 & 5 & 7 & 7 & 5 \\ 7 & 2 & 6 & 2 & 6 \\ 6 & 9 & 7 & 7 & 0 \\ 6 & 6 & 1 & 7 & 6 \\ 9 & 3 & 2 & 4 & 8 \end{bmatrix}$$

      i) Plot the histogram of the image.
      ii) Perform histogram equalization and then show the equalized image and plot histogram again.
      iii) Histogram matching is a useful contrast manipulation technique that transforms an image's histogram to match with that of another image. Clearly describe how can you achieve it.
   c) What do you mean by contrast stretching? Explain a simple non-linear contrast stretching method. (08)

7. a) Define full color image processing. Discuss about intensity slicing with appropriate diagram. (08)
   b) Define convex hull. Draw the convex hull of the following binary image. (17)



Binary imge: A



Structure elements: c is the center

   c) Define morphological gradient. When do we use top-hat and bottom-hat transformation in grayscale image? (10)

8. a) Define pigments. Write down the process of conversion of pixel value from RGB color space to HSI color space. (16)
   b) Prove that in continuous domain, probability distribution function of equalized image is always uniform. Discuss it with appropriate example. (07)
   c) Consider a portion of a binary image as shown below. Perform the following operations on the image. (12)
      i) Opening
      ii) Closing

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \end{bmatrix}$$

Binary image

[① 1 1]

Structuring Element (SE)

The circle indicates the center of SE